

Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

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Tuesday 21 April 2026

1.1 Daily Limit

Logarithmic Limit of a Power of the Sinc Function

Evaluate the limit:

$$\lim_{x \rightarrow 0} \left(\frac{\sin(x)}{x} \right)^{\frac{1}{x^2}}$$

Solution: Let L denote the limit. To transform this indeterminate exponential form of type 1^∞ into an evaluable algebraic structure, we take the natural logarithm of both sides:

$$\ln(L) = \lim_{x \rightarrow 0} \frac{1}{x^2} \ln \left(\frac{\sin(x)}{x} \right)$$

We begin by expressing $\sin(x)$ using its standard Maclaurin power series expansion around $x = 0$:

$$\sin(x) = x - \frac{x^3}{6} + \frac{x^5}{120} - O(x^7)$$

Dividing the entire series by x yields the power series for the sinc function:

$$\frac{\sin(x)}{x} = 1 - \frac{x^2}{6} + \frac{x^4}{120} - O(x^6)$$

Next, we substitute this series into the logarithmic expansion formula $\ln(1 - u) = -u - \frac{u^2}{2} - O(u^3)$, where $u = \frac{x^2}{6} - \frac{x^4}{120} + O(x^6)$:

$$\ln \left(\frac{\sin(x)}{x} \right) = - \left(\frac{x^2}{6} - \frac{x^4}{120} \right) - \frac{1}{2} \left(\frac{x^2}{6} \right)^2 + O(x^6)$$

Expanding and collecting like terms of order x^4 :

$$\ln \left(\frac{\sin(x)}{x} \right) = -\frac{x^2}{6} + \left(\frac{1}{120} - \frac{1}{72} \right) x^4 + O(x^6) = -\frac{x^2}{6} - \frac{x^4}{180} + O(x^6)$$

Substituting this expanded logarithmic expression back into our limit yields:

$$\ln(L) = \lim_{x \rightarrow 0} \frac{-\frac{x^2}{6} - \frac{x^4}{180} + O(x^6)}{x^2} = \lim_{x \rightarrow 0} \left(-\frac{1}{6} - \frac{x^2}{180} + O(x^4) \right) = -\frac{1}{6}$$

Exponentiating both sides to solve for L , we obtain the exact closed-form value:

$$L = e^{-\frac{1}{6}} = \frac{1}{\sqrt[6]{e}}$$

Wednesday 22 April 2026

2.1 Daily Limit

Asymptotic Exponential Limit of a Shifted Power

Evaluate the limit:

$$\lim_{x \rightarrow +\infty} \frac{\left(1 + \frac{1}{x}\right)^{x^2}}{e^x}$$

Solution: Let us define the function $f(x) = \frac{(1+1/x)^{x^2}}{e^x}$. To evaluate its asymptotic behavior as $x \rightarrow +\infty$, we take the natural logarithm:

$$\ln(f(x)) = \ln \left[\frac{\left(1 + \frac{1}{x}\right)^{x^2}}{e^x} \right] = x^2 \ln \left(1 + \frac{1}{x} \right) - x$$

As $x \rightarrow +\infty$, the term $u = \frac{1}{x} \rightarrow 0^+$. We apply the Taylor series expansion for the natural logarithm around zero:

$$\ln(1+u) = u - \frac{u^2}{2} + \frac{u^3}{3} - O(u^4)$$

Substituting $u = \frac{1}{x}$ into this expansion gives:

$$\ln \left(1 + \frac{1}{x} \right) = \frac{1}{x} - \frac{1}{2x^2} + \frac{1}{3x^3} - O \left(\frac{1}{x^4} \right)$$

Multiplying this entire series by the factor x^2 :

$$x^2 \ln \left(1 + \frac{1}{x} \right) = x - \frac{1}{2} + \frac{1}{3x} - O \left(\frac{1}{x^2} \right)$$

We now subtract the linear term x as required by our logarithmic expression:

$$\ln(f(x)) = \left(x - \frac{1}{2} + \frac{1}{3x} - O \left(\frac{1}{x^2} \right) \right) - x = -\frac{1}{2} + \frac{1}{3x} - O \left(\frac{1}{x^2} \right)$$

Taking the limit as $x \rightarrow +\infty$, all inverse power terms vanish identically:

$$\lim_{x \rightarrow +\infty} \ln(f(x)) = -\frac{1}{2}$$

By the continuity of the exponential function, we exponentiate to find the final limit:

$$L = \lim_{x \rightarrow +\infty} f(x) = \exp \left(-\frac{1}{2} \right) = \frac{1}{\sqrt{e}}$$

Thursday 23 April 2026

3.1 Daily Limit

Limit of an Exponentially Scaled Harmonic Integral

Evaluate the limit:

$$\lim_{n \rightarrow \infty} \frac{1}{n^2} \exp \left(4 \int_0^1 \frac{1 - x^n}{1 - x + x^2 - x^3} dx \right)$$

Solution: Let I_n denote the definite integral inside the exponential argument. We begin by algebraically factoring the cubic polynomial in the denominator by grouping:

$$1 - x + x^2 - x^3 = (1 - x) + x^2(1 - x) = (1 - x)(1 + x^2)$$

Thus, the integral can be rewritten as:

$$I_n = \int_0^1 \frac{1 - x^n}{(1 - x)(1 + x^2)} dx$$

We perform partial fraction decomposition on the rational function $\frac{1}{(1-x)(1+x^2)}$:

$$\frac{1}{(1-x)(1+x^2)} = \frac{A}{1-x} + \frac{Bx+C}{1+x^2}$$

Multiplying through by the common denominator yields $1 = A(1+x^2) + (Bx+C)(1-x)$. Setting $x = 1$ gives $2A = 1 \implies A = \frac{1}{2}$. Expanding and equating coefficients of x^2 and constant terms confirms that $B = \frac{1}{2}$ and $C = \frac{1}{2}$. Therefore:

$$\frac{1}{(1-x)(1+x^2)} = \frac{1}{2(1-x)} + \frac{x+1}{2(1+x^2)}$$

Multiplying this decomposition by $(1-x^n)$ and integrating over $[0, 1]$ separates I_n into three distinct components:

$$I_n = \frac{1}{2} \int_0^1 \frac{1 - x^n}{1 - x} dx + \frac{1}{2} \int_0^1 \frac{x+1}{1+x^2} dx - \frac{1}{2} \int_0^1 x^n \frac{x+1}{1+x^2} dx$$

We evaluate each integral individually: 1. The first integral recognizes the sum of a finite geometric progression:

$$\int_0^1 \frac{1 - x^n}{1 - x} dx = \int_0^1 \sum_{k=0}^{n-1} x^k dx = \sum_{k=1}^n \frac{1}{k} = H_n$$

where H_n is the n -th harmonic number, with asymptotic expansion $H_n = \ln(n) + \gamma + o(1)$ as $n \rightarrow \infty$, where γ is the Euler-Mascheroni constant. 2. The second integral evaluates to elementary constants:

$$\int_0^1 \frac{x+1}{1+x^2} dx = \left[\frac{1}{2} \ln(1+x^2) + \arctan(x) \right]_0^1 = \frac{1}{2} \ln(2) + \frac{\pi}{4}$$

3. The third integral vanishes asymptotically as $n \rightarrow \infty$ due to the uniform boundedness of $\frac{x+1}{1+x^2} \leq 2$ on $[0, 1]$:

$$0 \leq \int_0^1 x^n \frac{x+1}{1+x^2} dx \leq 2 \int_0^1 x^n dx = \frac{2}{n+1} \rightarrow 0$$

Combining these asymptotic results for I_n :

$$I_n = \frac{1}{2} \ln(n) + \frac{\gamma}{2} + \frac{1}{4} \ln(2) + \frac{\pi}{8} + o(1)$$

Multiplying by 4 as required by the exponent:

$$4I_n = 2 \ln(n) + 2\gamma + \ln(2) + \frac{\pi}{2} + o(1) = \ln(n^2) + \ln\left(2e^{2\gamma+\frac{\pi}{2}}\right) + o(1)$$

Substituting this back into our original limit expression:

$$L = \lim_{n \rightarrow \infty} \frac{1}{n^2} \exp(4I_n) = \lim_{n \rightarrow \infty} \frac{1}{n^2} \cdot n^2 \cdot 2e^{2\gamma+\frac{\pi}{2}} \cdot e^{o(1)} = 2e^{2\gamma+\frac{\pi}{2}}$$

Friday 24 April 2026

4.1 Daily Limit

Asymptotic Limit of a Triple Sum Involving Zeta and Gamma Functions

Evaluate the limit:

$$\lim_{x \rightarrow \infty} \sum_{k=0}^{2026} \sum_{n=1}^x \sum_{s=1}^x \frac{(-1)^{nk} \zeta(xnk)}{x^2 \Gamma(n)}$$

Solution: We observe that the innermost summation variable s appears strictly independently of the indices n and k . Therefore, we can factor the sum over s out of the double summation:

$$S(x) = \left(\frac{1}{x^2} \sum_{s=1}^x s \right) \left(\sum_{k=0}^{2026} \sum_{n=1}^x \frac{(-1)^{nk} \zeta(xnk)}{\Gamma(n)} \right)$$

We evaluate the algebraic limit of the first factor using the standard formula for the sum of the first x integers:

$$\lim_{x \rightarrow \infty} \frac{1}{x^2} \sum_{s=1}^x s = \lim_{x \rightarrow \infty} \frac{x(x+1)}{2x^2} = \frac{1}{2}$$

Thus, our problem reduces to evaluating the remaining sum as $x \rightarrow \infty$:

$$L = \frac{1}{2} \lim_{x \rightarrow \infty} \sum_{k=0}^{2026} \sum_{n=1}^x \frac{(-1)^{nk} \zeta(xnk)}{\Gamma(n)}$$

We analyze the behavior of the Riemann zeta term $\zeta(xnk)$ across two distinct cases for the index k : 1. **Case $k = 0$:** The argument of the zeta function becomes identically zero. Since $\zeta(0) = -\frac{1}{2}$ and $(-1)^0 = 1$, the sum over n yields:

$$\sum_{n=1}^{\infty} \frac{-1/2}{\Gamma(n)} = -\frac{1}{2} \sum_{n=1}^{\infty} \frac{1}{(n-1)!} = -\frac{1}{2} \sum_{m=0}^{\infty} \frac{1}{m!} = -\frac{e}{2}$$

2. **Case $1 \leq k \leq 2026$:** For any positive integers $n \geq 1$ and $k \geq 1$, the product $nk \geq 1$. As $x \rightarrow \infty$, the argument $xnk \rightarrow \infty$. Since $\lim_{y \rightarrow \infty} \zeta(y) = 1$ uniformly, we replace $\zeta(xnk)$ by 1:

$$\lim_{x \rightarrow \infty} \sum_{n=1}^x \frac{(-1)^{nk} \zeta(xnk)}{\Gamma(n)} = \sum_{n=1}^{\infty} \frac{(-1)^{nk}}{(n-1)!}$$

We separate this evaluation further based on the parity of k : * If k is **even**, $(-1)^{nk} = 1$ for all n , giving $\sum_{n=1}^{\infty} \frac{1}{(n-1)!} = e$. In the interval $[1, 2026]$, there are exactly 1013 even values of k . * If k is **odd**, $(-1)^{nk} = (-1)^n$, giving $\sum_{n=1}^{\infty} \frac{(-1)^n}{(n-1)!} = -\sum_{m=0}^{\infty} \frac{(-1)^m}{m!} = -e^{-1} = -\frac{1}{e}$. In the interval $[1, 2026]$, there are exactly 1013 odd values of k .

Summing all contributions across $0 \leq k \leq 2026$:

$$\sum_{k=0}^{2026} \lim_{x \rightarrow \infty} \sum_{n=1}^x \frac{(-1)^{nk} \zeta(xnk)}{\Gamma(n)} = -\frac{e}{2} + 1013e - \frac{1013}{e} = \frac{2025}{2}e - \frac{1013}{e}$$

Multiplying by the overall outer factor of $\frac{1}{2}$ yields the exact closed-form solution:

$$L = \frac{1}{2} \left(\frac{2025}{2}e - \frac{1013}{e} \right) = \frac{2025e}{4} - \frac{1013}{2e}$$

Saturday 25 April 2026

5.1 Daily Limit

Right-Hand Limit of a Difference of Radicals Involving Arccosine

Evaluate the limit:

$$\lim_{x \rightarrow -1^+} \frac{\sqrt{\pi} - \sqrt{\arccos(x)}}{\sqrt{x+1}}$$

Solution: We introduce the local coordinate substitution $x = -1 + t$, where $t \rightarrow 0^+$ as $x \rightarrow -1^+$. Our objective is to determine the asymptotic expansion of $\arccos(-1 + t)$ for small positive values of t .

Let $\theta = \arccos(-1 + t)$, which implies $\cos(\theta) = -1 + t$. Since t is positive and strictly near zero, θ is slightly less than π . We define the complementary angle $\phi = \pi - \theta$, where $\phi \rightarrow 0^+$ as $t \rightarrow 0^+$. Using the trigonometric identity $\cos(\pi - \phi) = -\cos(\phi)$:

$$-\cos(\phi) = -1 + t \implies \cos(\phi) = 1 - t$$

We expand $\cos(\phi)$ using its Maclaurin series around $\phi = 0$:

$$1 - \frac{\phi^2}{2} + O(\phi^4) = 1 - t \implies \frac{\phi^2}{2} = t + O(\phi^4)$$

Solving for ϕ , we take the principal square root:

$$\phi = \sqrt{2t} + O(t^{3/2})$$

Recalling that $\theta = \pi - \phi$, we obtain the expansion for the inverse cosine function near the left endpoint of its domain:

$$\arccos(-1 + t) = \pi - \sqrt{2t} + O(t^{3/2})$$

We now substitute this expansion into the inner radical of the numerator:

$$\sqrt{\arccos(-1 + t)} = \sqrt{\pi - \sqrt{2t} + O(t^{3/2})} = \sqrt{\pi} \left(1 - \frac{\sqrt{2t}}{\pi} + O(t^{3/2}) \right)^{\frac{1}{2}}$$

Using the binomial approximation $(1 - u)^{1/2} = 1 - \frac{1}{2}u + O(u^2)$ for small u :

$$\sqrt{\arccos(-1 + t)} = \sqrt{\pi} \left(1 - \frac{\sqrt{2t}}{2\pi} + O(t) \right) = \sqrt{\pi} - \sqrt{\frac{t}{2\pi}} + O(t)$$

Subtracting this result from $\sqrt{\pi}$ clearly isolates the leading-order behavior of the numerator:

$$\sqrt{\pi} - \sqrt{\arccos(-1 + t)} = \sqrt{\frac{t}{2\pi}} - O(t)$$

Dividing by the denominator $\sqrt{x+1} = \sqrt{t}$ and passing to the limit as $t \rightarrow 0^+$:

$$L = \lim_{t \rightarrow 0^+} \frac{\sqrt{\frac{t}{2\pi}} - O(t)}{\sqrt{t}} = \lim_{t \rightarrow 0^+} \left(\frac{1}{\sqrt{2\pi}} - O(\sqrt{t}) \right) = \frac{1}{\sqrt{2\pi}} = \frac{\sqrt{2\pi}}{2\pi}$$

Sunday 26 April 2026

6.1 Daily Limit

Limit of a Difference Quotient Involving a Logarithmic Shift

Evaluate the limit:

$$L = \lim_{x \rightarrow 0} \frac{f(x) - f(\ln(1+x))}{x^3}$$

where f is twice differentiable, $f'(0) = 0$, and $f''(0) = 4$.

Solution: Since the function f is twice differentiable at $x = 0$ with $f'(0) = 0$, we can write its local Peano Taylor expansion around zero as:

$$f(u) = f(0) + \frac{1}{2}f''(0)u^2 + Cu^3 + o(u^3) = f(0) + 2u^2 + Cu^3 + o(u^3)$$

where C is a constant representing the third-order behavior (if it exists, or treated algebraically via expansion order). Let us apply this expansion to evaluate $f(x)$:

$$f(x) = f(0) + 2x^2 + Cx^3 + o(x^3)$$

Next, we determine the expansion for the composite term $f(\ln(1+x))$. We first recall the Maclaurin series expansion for the logarithmic argument:

$$u = \ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - o(x^3)$$

We compute the required powers of u up to the third order: 1. For the quadratic term:

$$u^2 = \left(x - \frac{x^2}{2} + o(x^2)\right)^2 = x^2 - x^3 + o(x^3)$$

2. For the cubic term:

$$u^3 = (x - o(x))^3 = x^3 - o(x^3)$$

Substituting these powers back into our general polynomial expansion for $f(u)$:

$$\begin{aligned} f(\ln(1+x)) &= f(0) + 2(x^2 - x^3 + o(x^3)) + C(x^3 - o(x^3)) + o(x^3) \\ &= f(0) + 2x^2 + (C-2)x^3 + o(x^3) \end{aligned}$$

We now subtract $f(\ln(1+x))$ from $f(x)$. Notice how the constant term $f(0)$, the quadratic term $2x^2$, and the arbitrary cubic coefficient C all cancel out identically:

$$\begin{aligned} f(x) - f(\ln(1+x)) &= [f(0) + 2x^2 + Cx^3] - [f(0) + 2x^2 + (C-2)x^3] + o(x^3) \\ &= 2x^3 + o(x^3) \end{aligned}$$

Substituting this difference back into the quotient limit:

$$L = \lim_{x \rightarrow 0} \frac{2x^3 + o(x^3)}{x^3} = \lim_{x \rightarrow 0} (2 + o(1)) = 2$$

Thus, the limit evaluates exactly to $\frac{1}{2}f''(0) = 2$.

Monday 27 April 2026

7.1 Daily Limit

Asymptotic Limit of a Shifted Tangent Difference

Evaluate the limit:

$$\lim_{x \rightarrow \infty} x^2 \left(\tan \left(\frac{1}{x} \right) - \tan \left(\frac{1}{x+5} \right) \right)$$

Solution: To analyze the asymptotic behavior as $x \rightarrow \infty$, we utilize the trigonometric difference identity for tangents:

$$\tan(A) - \tan(B) = \frac{\sin(A - B)}{\cos(A) \cos(B)}$$

Setting $A = \frac{1}{x}$ and $B = \frac{1}{x+5}$, the argument of the sine function in the numerator simplifies over a common denominator:

$$A - B = \frac{1}{x} - \frac{1}{x+5} = \frac{(x+5) - x}{x(x+5)} = \frac{5}{x^2 + 5x}$$

Substituting this algebraic identity into our difference expression transforms the limit into:

$$L = \lim_{x \rightarrow \infty} x^2 \left[\frac{\sin \left(\frac{5}{x^2 + 5x} \right)}{\cos \left(\frac{1}{x} \right) \cos \left(\frac{1}{x+5} \right)} \right]$$

We evaluate the denominator independently using continuity: as $x \rightarrow \infty$, both arguments $\frac{1}{x} \rightarrow 0$ and $\frac{1}{x+5} \rightarrow 0$. Therefore:

$$\lim_{x \rightarrow \infty} \cos \left(\frac{1}{x} \right) \cos \left(\frac{1}{x+5} \right) = \cos(0) \cos(0) = 1 \cdot 1 = 1$$

For the numerator, we multiply and divide by the algebraic argument of the sine function to construct the fundamental trigonometric limit structure:

$$L = \lim_{x \rightarrow \infty} \left[\frac{\sin \left(\frac{5}{x^2 + 5x} \right)}{\frac{5}{x^2 + 5x}} \cdot \frac{5x^2}{x^2 + 5x} \cdot \frac{1}{\cos \left(\frac{1}{x} \right) \cos \left(\frac{1}{x+5} \right)} \right]$$

Since $\theta = \frac{5}{x^2 + 5x} \rightarrow 0$ as $x \rightarrow \infty$, the standard limit $\lim_{\theta \rightarrow 0} \frac{\sin(\theta)}{\theta} = 1$ applies directly. Evaluating the rational function factor by dividing numerator and denominator by x^2 :

$$\lim_{x \rightarrow \infty} \frac{5x^2}{x^2 + 5x} = \lim_{x \rightarrow \infty} \frac{5}{1 + \frac{5}{x}} = 5$$

Combining all evaluated limits into the product:

$$L = 1 \cdot 5 \cdot 1 = 5$$

Tuesday 28 April 2026

8.1 Daily Limit

Asymptotic Evaluation of an Exponential and Trigonometric Limit

Evaluate the limit:

$$\lim_{x \rightarrow 0} \frac{(e^x - 1)^2 (\ln(1 + x) - \sin(x))^2}{x^n} = \alpha$$

If α is a finite and non-zero real number, find the value of $2\alpha n$.

Solution: For the limit α to converge to a finite, non-zero real number, the polynomial degree n of the denominator must match the exact leading asymptotic order of the numerator as $x \rightarrow 0$. We determine this order by expanding each factor into its Maclaurin power series around $x = 0$.

1. **The Exponential Factor:** We use the standard linear approximation $e^x - 1 = x + O(x^2)$. Squaring this expression yields:

$$(e^x - 1)^2 = (x + O(x^2))^2 = x^2 + O(x^3)$$

2. **The Logarithmic and Trigonometric Factor:** We expand both functions up to their cubic terms:

$$\ln(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - O(x^4)$$

$$\sin(x) = x - \frac{x^3}{6} + O(x^5)$$

Subtracting the sine series from the logarithm series cancels the linear x term:

$$\ln(1 + x) - \sin(x) = \left(x - \frac{x^2}{2} + \frac{x^3}{3}\right) - \left(x - \frac{x^3}{6}\right) + O(x^4) = -\frac{x^2}{2} + \frac{1}{2}x^3 + O(x^4)$$

Squaring this difference isolates the dominant quadratic term:

$$(\ln(1 + x) - \sin(x))^2 = \left(-\frac{x^2}{2} + O(x^3)\right)^2 = \frac{x^4}{4} + O(x^5)$$

Multiplying the two simplified asymptotic factors together gives the leading behavior of the entire numerator:

$$(e^x - 1)^2 (\ln(1 + x) - \sin(x))^2 = (x^2 + O(x^3)) \left(\frac{x^4}{4} + O(x^5)\right) = \frac{x^6}{4} + O(x^7)$$

Substituting this asymptotic expansion back into the original limit quotient:

$$\alpha = \lim_{x \rightarrow 0} \frac{\frac{x^6}{4} + O(x^7)}{x^n}$$

In order for this limit to equal a finite, non-zero constant α , the power of x in the denominator must equal the degree of the leading term in the numerator. Therefore, we immediately deduce:

$$n = 6 \implies \alpha = \frac{1}{4}$$

We are asked to compute the final combined value of the expression $2\alpha n$:

$$2\alpha n = 2 \cdot \left(\frac{1}{4}\right) \cdot 6 = \frac{12}{4} = 3$$

Wednesday 29 April 2026

9.1 Daily Limit

Limit of a Scaled Multivariable Integral of a Squared Sum

Evaluate the limit:

$$\lim_{n \rightarrow \infty} \frac{1}{n^2} \int_0^1 \cdots \int_0^1 \left(\sum_{j=1}^n x_j \right)^2 dx_1 \cdots dx_n$$

Solution: Let I_n denote the n -fold multivariable integral over the unit hypercube $[0, 1]^n$. We begin by algebraically expanding the integrand using the multinomial theorem:

$$\left(\sum_{j=1}^n x_j \right)^2 = \sum_{j=1}^n x_j^2 + 2 \sum_{1 \leq i < j \leq n} x_i x_j$$

By linearity of integration, we can distribute the multiple integrals across each individual term in the expansion:

$$I_n = \sum_{j=1}^n \left(\int_0^1 \cdots \int_0^1 x_j^2 dx_1 \cdots dx_n \right) + 2 \sum_{1 \leq i < j \leq n} \left(\int_0^1 \cdots \int_0^1 x_i x_j dx_1 \cdots dx_n \right)$$

Since the variables $x_1, \dots, x_n$ are strictly independent, each multivariable integral separates into a product of one-dimensional integrals over $[0, 1]$: 1. For each squared term x_j^2 , the integration with respect to x_j yields $\int_0^1 x_j^2 dx_j = \frac{1}{3}$, while the remaining $n - 1$ integrals over dx_k ($k \neq j$) each evaluate to 1. There are exactly n such identical terms:

$$\sum_{j=1}^n \frac{1}{3} = \frac{n}{3}$$

2. For each cross-product term $x_i x_j$ ($i \neq j$), the integrations with respect to x_i and x_j yield $\left(\int_0^1 x_i dx_i \right) \left(\int_0^1 x_j dx_j \right) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$, while the remaining $n - 2$ integrations evaluate to 1. The number of distinct pairs (i, j) with $i < j$ is given by the binomial coefficient $\binom{n}{2} = \frac{n(n-1)}{2}$:

$$2 \sum_{1 \leq i < j \leq n} \frac{1}{4} = 2 \cdot \frac{n(n-1)}{2} \cdot \frac{1}{4} = \frac{n(n-1)}{4}$$

Summing these two components gives the exact polynomial formula for I_n :

$$I_n = \frac{n}{3} + \frac{n^2 - n}{4} = \frac{n^2}{4} + n \left(\frac{1}{3} - \frac{1}{4} \right) = \frac{n^2}{4} + \frac{n}{12}$$

We now substitute this closed-form integral value back into the limit expression and divide by n^2 :

$$L = \lim_{n \rightarrow \infty} \frac{1}{n^2} \left(\frac{n^2}{4} + \frac{n}{12} \right) = \lim_{n \rightarrow \infty} \left(\frac{1}{4} + \frac{1}{12n} \right) = \frac{1}{4}$$

Thursday 30 April 2026

10.1 Daily Limit

Limit of an Algebraic and Trigonometric Difference Quotient

Evaluate the limit:

$$\lim_{x \rightarrow 0} \frac{\sqrt[3]{1 + \sin^2(x)} - \sqrt{\cos(x)}}{x^2}$$

Solution: To evaluate this indeterminate limit of form $\frac{0}{0}$ cleanly without tedious multiple differentiations, we apply Maclaurin series expansions around $x = 0$ up to the second order.

1. **The Cube Root Term:** We first recall the series for the square of the sine function:

$$\sin(x) = x - O(x^3) \implies \sin^2(x) = x^2 - O(x^4)$$

We substitute this into the generalized binomial expansion $(1 + u)^{1/3} = 1 + \frac{1}{3}u - O(u^2)$, where $u = \sin^2(x)$:

$$\sqrt[3]{1 + \sin^2(x)} = \left(1 + x^2 - O(x^4)\right)^{\frac{1}{3}} = 1 + \frac{1}{3}x^2 - O(x^4)$$

2. **The Square Root Term:** We recall the Maclaurin expansion for the cosine function:

$$\cos(x) = 1 - \frac{x^2}{2} + O(x^4)$$

We substitute this into the binomial expansion $(1 - u)^{1/2} = 1 - \frac{1}{2}u - O(u^2)$, where $u = \frac{x^2}{2}$:

$$\sqrt{\cos(x)} = \left(1 - \frac{x^2}{2} + O(x^4)\right)^{\frac{1}{2}} = 1 - \frac{1}{2}\left(\frac{x^2}{2}\right) - O(x^4) = 1 - \frac{1}{4}x^2 - O(x^4)$$

We now subtract the square root expansion from the cube root expansion in the numerator. Notice how the leading constant terms cleanly cancel:

$$\begin{aligned} \sqrt[3]{1 + \sin^2(x)} - \sqrt{\cos(x)} &= \left(1 + \frac{1}{3}x^2\right) - \left(1 - \frac{1}{4}x^2\right) + O(x^4) \\ &= \left(\frac{1}{3} + \frac{1}{4}\right)x^2 + O(x^4) = \frac{7}{12}x^2 + O(x^4) \end{aligned}$$

Substituting this simplified quadratic difference back into our limit expression and dividing by x^2 :

$$L = \lim_{x \rightarrow 0} \frac{\frac{7}{12}x^2 + O(x^4)}{x^2} = \lim_{x \rightarrow 0} \left(\frac{7}{12} + O(x^2)\right) = \frac{7}{12}$$